AI 101 for Mechanical Engineers · Module 2

Physics, data and hybrid models

Lesson 2.1

What you will learn

> Compare physics-based, data-driven and hybrid models, and use the pump example to distinguish hydraulic power, input power and a residual correction.

Lesson overview

Module 2 · Lesson 1

Physics, data and hybrid models

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1 Physics-based model

Uses physical relationships and stated assumptions.

Inputs → Physical calculation → Prediction

Check assumptions against measurements.

2 Data-driven model

Learns input–output relationships from observations.

Operating records → Learned model → Prediction

Needs observations that cover the intended use.

3 Hybrid model

Combines a physical baseline with a learned correction.

Physical baseline + Learned correction → Corrected prediction

Useful when residuals show a repeatable pattern.

Pump-power example

$$P_h = \Delta p \times Q$$

P_h: hydraulic power (W) · Δp: pressure rise (Pa) · Q: volumetric flow rate (m³/s)

Given: Δp = 100 kPa = 100,000 Pa · Q = 0.02 m³/s

$$P_h = 100{,}000\ \text{Pa} \times 0.02\ \text{m}^3/\text{s} = 2{,}000\ \text{W}$$

$$P_{in} = \frac{P_h}{\eta} = \frac{2{,}000\ \text{W}}{0.8} = 2{,}500\ \text{W}$$

P_{in}: input power (W) · η: efficiency (dimensionless)

Hydraulic, shaft and electrical power are different. Use the efficiency that matches the quantities.

Residual = measured − predicted

$$r = y - \hat{y}$$

y: measured input power · $\hat{y}$: predicted input power · r: residual (W)

$$r = 2{,}800\ \text{W} - 2{,}500\ \text{W} = +300\ \text{W}$$

The prediction is 300 W below the measurement.

A residual shows a difference. It does not prove the physical cause.

Key takeaway — **Check each model within the conditions supported by its assumptions and evidence.**

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Physics-based models use governing relationships

- Uses governing relationships, stated parameters and appropriate conditions [14]
- Pump: hydraulic power from pressure rise and flow rate [15]
- Still check it against assumptions and physical measurements

Data-driven and hybrid models learn differently

Data-driven

- Estimates input-output relationship from observations [16]
- e.g. electrical power from flow, pressure, speed
- No hydraulic-power equation explicitly imposed

Hybrid

- Combines physical knowledge with learning from data [14], [17]
- Here: physical baseline plus learned residual correction
- Useful when informative physics misses some observed effects

> Other hybrids differ, e.g. physical constraints in the training objective (Module 16).

Hydraulic power is pressure rise times flow

$$P_h = \Delta p \times Q$$

$$P_h = 100{,}000\ \text{Pa} \times 0.02\ \text{m}^3/\text{s} = 2{,}000\ \text{W}$$

- P_h in W; Δp in Pa; Q in m^3/s [15]
- Δp = 100 kPa = 100,000 Pa

Input power depends on the stated efficiency

$$P_{in} = P_h / \eta$$

- η is dimensionless; with $\eta = 0.8$, $P_{in} = 2{,}500$ W
- Match the efficiency to the stated input-power meaning
- Hydraulic, shaft and electrical power are different quantities

A residual shows a discrepancy, not its cause

$$r = y - \hat{y}$$

$$r = 2{,}800\ \mathrm{W} - 2{,}500\ \mathrm{W} = +300\ \mathrm{W}$$

- y: measured input power; ŷ: predicted input power
- Positive residual: baseline prediction is below the measurement
- It does not, by itself, prove the physical cause

Choose an approach within its evidence limits

Approach	Consider when
Physics-based	Relationships, parameters and assumptions suit the question
Data-driven	Representative observations; relationship hard to specify directly
Hybrid	Useful baseline; repeatable residual patterns; adequate data

> None guarantees reliable extrapolation; outside the validated evidence, check the model.

CHECK YOUR UNDERSTANDING

Why are 2,000 W and 2,500 W different in this example?

Answer before moving on.

CHECK YOUR UNDERSTANDING

Why are 2,000 W and 2,500 W different in this example?

> 2,000 W is hydraulic power. 2,500 W is input power, calculated using the stated dimensionless efficiency of 0.8.

Key takeaways

- Physics models use physical relationships
- Data-driven models learn from observations; hybrid models use both
- State the power quantity and use consistent SI units
- A residual can support a correction but does not establish cause

References

[14] J. Willard, X. Jia, S. Xu, M. Steinbach, and V. Kumar, “Integrating scientific knowledge with machine learning for engineering and environmental systems,” ACM Comput. Surv., vol. 55, no. 4, pp. 1–37, 2023. doi: 10.1145/3514228

[15] J. F. Gülich, Centrifugal Pumps, 3rd ed. Berlin, Germany: Springer, 2014. doi: 10.1007/978-3-642-40114-5

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[17] A. Karpatne et al., “Theory-guided data science: A new paradigm for scientific discovery from data,” IEEE Trans. Knowl. Data Eng., vol. 29, no. 10, pp. 2318–2331, 2017. doi: 10.1109/TKDE.2017.2720168

Numbers match the course reference list. The written lesson is the authoritative text.